Blackwish Protocol Vehicle Operation Research

Representative land air water and space systems

Research reference dated 22 October 2026

RESEARCH / REVIEWED / NOT RATIFIED
REVIEWED means internal text-only desk review. It does not mean certification, training completion, regulatory acceptance, canonical adoption or approval to operate.

Vehicle operation is more than moving familiar controls. An input changes forces or a commanded state; the environment and vehicle dynamics determine the response; instruments and observation reveal what actually happened. Effective operation also depends on route information, coordinated roles, resources, maintenance and a suitable end state.

This reference compares representative land, air, water and space families. It explains ordinary conceptual controls, navigation and phase differences, then applies those distinctions to fictional vehicles and Blackwish consistency review. It is a breadth map with a method for researching an omitted type, not a catalogue of every vehicle or model.

How to use this reference

Understand the medium. Read the domain sections for the connection between input, motion and feedback. **Compare the operating system.** Use the navigation, crew, access and training sections. **Examine a claim.** Fill the vehicle card, keep fiction separate from actual-world evidence, then use the three Zen reviews and bounded Death Knell audit.

Scope and authority

This additive research artifact changes no canonical text or live system and reports no vehicle operation or experiment on people. It provides no ready-to-use startup checklist, emergency maneuver, control setpoint, loading limit or mission plan. Exact manuals, configuration, qualified instruction and applicable authority govern real operation.

Manual ratification is still required for a proposed Blackwish revision. Ratifying research does not itself authorize real vehicle operation or deployment; those require separate explicit action authorization and applicable qualifications.

Evidence checked: 22 October 2026. Source identifiers appear beside factual material; exact clickable URLs are in the register. Publication dates are stated only where established. A check date is not a publication date.

A shared model of operation

A useful conceptual loop is **intent and authority -> control input -> physical response -> sensed state -> interpretation and adjustment**. It is a way to ask questions, not an operating sequence. Control shape is weak evidence: a joystick can command a tool, thrust, rotation or automated position. Verify its mapping and mode. [L01, L02]

Separate function	Question the research must answer
Support and propulsion	What supports the vehicle, and what produces motion? A wing, wheel, buoyant hull, foil and rocket do different jobs.
Control and feedback	Which input changes which state? What lag, coupling and limits intervene? What shows actual state rather than requested state?
Navigation and guidance	How is position or motion estimated? What path is desired, physically feasible and authorized?
Resources and coordination	What is consumed, replenished or shared? Which roles and external infrastructure must remain available?
Completion and recovery	What does arrival, parking, mooring, landing, recovery or safe mode mean for this particular system?

Lifecycle questions across domains

Conceptual phase	Main comparison
Authorize fit and plan	Purpose, role competence, access, permission, environment, route and reserves.
Inspect load and brief	Actual configuration, loading, maintenance status, people, communications and readiness.
Depart or launch	A phase transition can change control authority, support and available recovery choices.
Navigate and manage	Monitor motion, traffic, energy, people, information quality and changing conditions.
Arrive or recover	Reach the appropriate end state; do not assume one universal meaning of stop.
Secure inspect and record	Secure by the applicable type procedure, report defects, replenish and support maintenance.

Lifecycle synthesis informed by the domain sources. Maintenance completion does not make preflight inspection unnecessary. A loss of propulsion or communications does not erase momentum, environmental forces or onboard safety duties. [P2, P3, F3]

Road and rail operation

Cycles and motorcycles

Pedaling supplies drive torque; steering, balance and braking interact through wheel contact and rider position. Cargo cycles, tricycles, recumbents and handcycles have different handling and fit. Electric assistance adds energy and mode management; legal class and path access depend on jurisdiction. [L03]

Typical motorcycles combine throttle, front and rear brakes, and clutch and gear selection where fitted. A two-wheeler requires balance and lean management; sidecars and three-wheelers are not interchangeable handling models. Feedback includes speed, gear, engine or battery state, road position and direct sensory information. Knowing a handbook is not the practical skill of controlling direction and speed. [L04]

Cars vans trucks and buses

Accelerator demand changes driving force; steering changes tire forces and curves the path. Friction brakes dissipate kinetic energy; regenerative braking can recover some energy where fitted. A pedal position does not guarantee a speed: mass, gradient, traction and resistance affect the response. Selected drive or gear, actual speed, fuel or charge and warning indications help interpret what the vehicle is doing. [L05, L06]

Departure, traffic flow, low-speed maneuvering and parking require different observation and space planning. Trucks, buses and trailers add length, offtracking, load or passenger management and sometimes air-brake systems. Familiar pedals or a car license do not establish competence with a heavy combination. The California commercial process separately assesses inspection, basic control and road skills; this is a US state example. [L06, L07]

Rail vehicles

A rail operator usually regulates longitudinal motion rather than freely steering. Traction and braking commands must be interpreted against route, grade, curvature, train length, coupler slack and distributed train forces. Cab information, signals, location cues and crew communications complement one another. A long linked train is not an isolated point responding instantly to a lever. [L08]

Signals, switches and movement authority govern where and when a train may proceed. Preparation, authorized departure, grade or curve management, station or yard approach and securement are distinct phases. Positive Train Control addresses specified hazards; it is not every operator duty or a universal obstacle detector. Mainline rules must not be imported wholesale into tram or isolated metro operation. [L08, L09]

Passenger versus operator: using a bus, tram or train as a passenger requires boarding, access and emergency-information awareness, not the professional qualifications needed to operate that system. These are different knowledge tasks.

Work vehicles and land edge families

Off road agricultural and construction machines

Differential track or wheel speeds can turn a skid-steer architecture; other machines use articulated or conventional steering. Travel commands move the chassis, while hydraulic or implement commands move a tool or load. The mapping may be configurable. A lifted load or attachment can alter stability as the work proceeds. [L01, L11, L13]

Feedback includes surroundings, attachment position, load indications where fitted, interlocks and warnings. Transit, approach, work cycle and securement are different modes; a ground worker's position and agreed signals are part of the operating picture. HSE distinguishes active-rider quad bikes from side-by-side vehicles, so one technique or restraint assumption cannot cover both. [L10]

Maintenance adds hazards from rotating shafts, hydraulics, raised equipment and stored energy. Engine stopped is not a general proof of isolation. Manufacturer procedures, appropriate safe-stop measures and qualified servicing remain type-specific. [L12, L13]

Human fit and mobility

Manual or powered wheelchairs, mobility scooters and handcycles deserve assessment on their own terms, not as small cars. Fit, transfer, reach, control mapping, route surfaces, battery or effort demands and access are specific to the person and device. Wheelchair slope or curb limits belong to the actual manual, not a universal number. Adapted road vehicles may use individualized controls and qualified driver-rehabilitation assessment. [L28, L29]

Other land families

Animal-drawn vehicles: reins, trained cues, harness and braking interact with a sentient animal, not a deterministic engine. Competence includes behavior, fit, suitable workload, rest and humane care. UK guidance treats horse preparation, vehicle condition and post-drive care separately. [L26, L27]

Cableways and lifts: this conceptual category is infrastructure-guided and often centrally controlled. Loading, doors, interlocks, station interfaces, drive and brake condition and evacuation arrangements replace free-steering concerns. UK frameworks recognize distinct regulated installations; no installation-specific operator manual was audited here. [L30, L31]

Snow and ice: runners, skis, tracks and iceboats extend the taxonomy, but surface strength, friction, exposure and terrain change the task. Their exact handling was not independently verified in this pass. Dedicated snowmobile and avalanche resources support the need for specialized training rather than extrapolation from roads. [L34]

Amphibious road craft: land, water and transition are separate modes. Configuration, stability, propulsion, loading and egress must each remain justified; a transition is not merely driving onward into water. [L32, L33]

Powered aircraft

Fixed wing aircraft

Wings provide aerodynamic lift; propulsion replenishes energy lost to drag. Typical controls connect ailerons with roll, elevator or stabilator with pitch, and rudder with yaw. Power, trim, flaps, spoilers and landing gear have design-specific roles. A normal turn involves bank and coordinated controls rather than car-like rudder steering. [A1, A2]

The operator relates control input to attitude, airspeed, altitude, vertical motion, heading and actual track. Airspeed describes motion through the air mass; groundspeed describes motion over the surface. Wind separates heading from ground track. Mass, centre of gravity, angle of attack, load factor and atmospheric conditions limit capability; an impressive instrument display does not enlarge those limits. [A1]

Preparation and ground movement, takeoff and climb, en-route flight, approach and landing, and shutdown are qualitatively different phases. Piston, turboprop, jet, electric and multiengine aircraft are not interchangeable operating types. A general description does not supply their limitations or procedures. [A1, A2]

Helicopters and gyroplanes

Helicopter rotor thrust supports and moves the craft. **Cyclic** changes rotor-disc orientation; **collective** changes blade pitch together; **antitorque controls** manage heading and torque balance by configuration-dependent means. Rotor speed and power management are central, while hover, transition and forward flight have different demands. Inputs are coupled rather than independent car-like commands. [A3]

A gyroplane is different: its rotor normally autorotates and a separate propeller provides forward thrust. It is not simply a helicopter with the engine off. FAA's catalogue warns that the older Rotorcraft Flying Handbook is the gyroplane reference, not current helicopter guidance. [A4]

Powered lift

Powered-lift designs combine vertical-lift and wing-borne phases, with diverse propulsion and control arrangements. Transition envelope, hover demand, landing-site conditions and mode awareness matter. A control inceptor may command different functions through automation as phase changes. FAA's US framework requires individual powered-lift type ratings; that does not establish one universal VTOL interface or worldwide qualification rule. [A9]

Feedback matters: the commanded attitude, selected automation mode, sensed aircraft state and resulting trajectory are not identical facts. A credible explanation keeps them separate.

Other aircraft and remote flight

Gliders

Gliders trade altitude for airspeed and reach, and can gain height in rising air. Aerodynamic controls coexist with drag devices such as spoilers or airbrakes. Launch may involve a tow aircraft, winch or installed motor. The operating picture includes launch coordination, weather and lift, energy state and remaining landing reach; a powered go-around must not be assumed. [A5]

Balloons and airships

Balloons use buoyancy rather than wing lift. Hot-air versions alter envelope temperature; gas balloons manage gas and ballast within their design. Horizontal travel largely follows the wind. Selecting altitude can exploit different winds but does not promise arbitrary steering. Thermal response, resources, landing areas and recovery crew belong in the model. [A6]

Airships combine buoyant support with propulsion and directional controls. Loading and trim, gas volume or pressure, wind, mooring and ground handling all matter. Ballonets in nonrigid designs manage envelope pressure and trim, not universal magical lift. FAA's linked airship manual is from 1942: useful historical principles, not a modern operating checklist. [A4, A7]

Uncrewed aircraft

Uncrewed is a crew arrangement, not an aerodynamic family. A UAS may be fixed-wing, rotorcraft, multirotor or hybrid. Operation includes the aircraft, control station, command link, navigation sensors and automation modes. Manual, assisted and waypoint modes and lost-link behavior are model- and configuration-dependent. [A8]

A remote operator needs evidence of actual aircraft state and a clear understanding of what is being commanded. An automated return function does not establish that a route, airspace or landing site is safe. US FAA remote-pilot rules are one jurisdictional example; actual permissions depend on the operation and authority. [A8]

Water and snow interfaces

Seaplanes combine water displacement or planing with aerodynamic flight. Water movement, takeoff transition, landing, mooring and surface conditions create tasks absent from a land-only airplane. Waves, wind, obstacles and configuration matter. Amphibians, skiplanes and float or ski helicopters require dedicated instruction rather than an inferred ability to operate on any surface. [A10]

Transition test: if support, control effect or available recovery changes, treat the transition as its own research problem. The capability to travel in two media is not evidence that every transition between them is usable.

Surface watercraft

Paddle oar and pedal craft

Human effort produces thrust; asymmetric strokes, oars, rudders or skegs where fitted and body or hull position influence course. Canoes, kayaks, rowing shells, rafts and paddleboards are distinct handling families. Crew synchronization can itself be a control function. Wind, current, fatigue, flotation, exposure and reachable landing points shape the journey. Sheltered-water familiarity is not sea-kayak rescue competence. [L14, L15]

Sailing craft

Sails produce aerodynamic force, while a hull, keel, centreboard or other lateral-resistance surface constrains sideways movement. The helm changes heading; sheets change sail setting; reefing changes the sail configuration. Steering, sail trim and crew balance form a coupled loop around course relative to wind. Conventional sailboats cannot sail directly into the wind, so an upwind destination requires an indirect course. [L16]

Feedback includes sail behavior, heel, wind, heading and speed. Apparent wind is the wind experienced by the moving craft; a calculated true-wind value depends on other inputs. The helm and sail handlers coordinate around maneuvers and crew positions. A generic points-of-sail diagram is not a performance guarantee for every rig or foiling craft. [L16, L36]

Powered boats and ships

Propellers, waterjets and other thrusters create thrust. Rudders, steerable drives, jets and differential thrust create turning moments, with different low-speed authority. Current and wind continue acting when propulsion changes; rotation, sideways drift and forward motion can coexist. Berthing is a positioning problem in a moving medium, not road parking on water. [L17, L35]

Observe actual heading, track and speed rather than treating helm position as the result. Compass, speed log, GNSS or chartplotter, depth, machinery indications and radar or AIS where fitted provide different information. Depth indication depends on sensor reference and settings and must not be confused with chart datum. [L18, L35, L36]

Departure and berthing require coordination of lookout, line handling and helm or engine orders. Underway, route monitoring and traffic appraisal dominate. Larger ships distribute work among bridge, engineering and deck teams; assignment depends on the vessel and operation. Canadian pleasure-craft guidance emphasizes passenger briefing, predeparture readiness and maintenance, not just control manipulation. [L17]

No unconditional right of way: lookout, collision-risk appraisal and the applicable navigation rules remain relevant. International and US Inland rules differ in specified respects; identify the actual rule set. [L19]

Specialized marine systems

Hydrofoils and hovercraft

Hydrofoils develop water-flow-dependent lift. Moving between hull-supported and foil-supported states changes stability and control. Candela is a specific example in which sensors and a flight controller adjust submerged foils; it does not prove all foils are automatically stabilized. Its owner's manual describes a particular wheel, throttle and touchscreen interface, not a universal marine cockpit. [L20, L37]

Hovercraft use an air cushion for support, with propulsion and steering systems for motion. Cushion state, lateral drift, surface changes and land-water transitions require type-specific handling. The UK hovercraft code includes control and failure training, limits and evacuation arrangements. IMO treats hydrofoils and air-cushion vehicles within its scope-dependent high-speed-craft framework. [L21, L22, L23]

Submarines and crewed submersibles

Separate buoyancy and trim, propulsion and attitude, navigation, pressure integrity, energy and life support. Large submarines and small research vehicles are not operationally interchangeable. Alvin is one documented example: variable ballast supports buoyancy management; electrical thrusters and digital controls support three-axis motion. Doppler velocity, gyro and acoustic information contribute to navigation. [L02, L25]

The pilot relates inputs to depth, position, bottom clearance and resources. Launch, descent, work at depth, ascent and recovery are distinct phases involving science planning, equipment checks, sign-offs, surface control and specialized support personnel. Alvin uses underwater telephone communication during dives and marine VHF at the surface, illustrating why underwater communication is not ordinary surface radio. [L02, L24]

Remote and autonomous underwater craft

A remotely operated underwater vehicle links vehicle functions with a topside control team and, often, a tether. An autonomous underwater vehicle combines mission definition with onboard autonomy. These are conceptual family distinctions here, not source-verified procedures for a particular ROV or AUV. Link, energy, navigation uncertainty and recovery arrangements need a dedicated card.

Navigation and recovery

Heading is not track; speed through water is not speed over ground. Charted depth, current water depth, vessel draft and available clearance must not be collapsed into one number. NOAA Coast Pilot complements charts with local channels, currents, hazards and facilities. Charts, notices, weather, tide information and lookout answer different questions. [L18]

Research boundary: stop, hold, surface, recover and abandon are not interchangeable actions. Do not invent a universally safe maneuver, reachable rescuer or available recovery mode to complete a description.

Spacecraft and planetary vehicles

Launch and orbital motion

Rocket thrust changes velocity while gravity, aerodynamic loads and changing mass shape ascent. Launch integrates preparation, powered ascent, separation and injection with flight computers, ground teams and range-safety roles. Propellant, available velocity change, loads and mission windows constrain the task; distance to the destination alone is insufficient. [S1, S2]

An orbiting spacecraft can coast without continuous propulsion. Engines change trajectory; they do not simply sustain road-like cruising speed. Attitude is pointing, while translation concerns position and velocity. Thrusters or momentum-exchange devices can control attitude; reaction wheels do not provide sustained translational propulsion. Rendezvous requires matching position and velocity, not merely pointing at a target. [S3, S5, F3]

Frames instruments and control

Navigation estimates state; guidance chooses the desired path; control commands effectors. Specify inertial, rotating planetary, body, local or target-relative frames and consistent time and units. A command sensible in one frame may mean something different in another. Telemetry, tracking and onboard sensors support estimates rather than perfect omniscient knowledge. [S3, S4]

Entry and landing

Atmospheric entry reintroduces aerodynamic forces and major thermal and structural constraints. Attitude, thermal protection and deceleration belong to a mission-designed entry before the type's landing system. The historical Shuttle used orbital thrusters and atmospheric surfaces in different regimes and landed as a glider. Neither its profile nor Orion's is universal. [S6, S7]

Probes landers and rovers

Deep-space teams send validated sequences and interpret telemetry; onboard autonomy handles time-critical work and faults. Pointing can couple communications, solar power, temperature and science. Light-time and interrupted contact make this unlike instantaneous joystick control. NASA's Mars communications paper is a 2023 architecture concept review, not evidence of flown crewed Mars operations. [S3, S5, P3]

Rovers add wheel-terrain interaction, slip, rocks, slopes, localization uncertainty and energy or thermal limits. Perseverance AutoNav can choose a path around obstacles toward assigned destinations; ground teams retain objective-setting and planning roles. Bounded driving autonomy is not unrestricted mission authority. Landing methods likewise depend on local atmosphere, gravity and terrain. [S9]

Resources are distinct: electrical power is not propellant. Crewed life support is not oxygen alone; it includes atmosphere and contaminant control, ventilation, temperature, water, waste and fire protection. [S10, S11]

Comparing control and navigation

The same words can hide different state variables. Use the table as a comparison aid, not as an operating instruction. The mechanisms are synthesized from the domain sources. [L05, L08, L18, A1, S3, S4]

Domain	Input and response	Feedback and navigation
Road	Steering and tire forces curve a path; drive and braking forces change speed within traction limits.	Actual speed, position, road users and selected mode; maps do not establish current clearance or permission.
Rail	Traction and braking act on a mechanically linked train; route geometry is infrastructure-guided.	Speed trend, location, signals, movement authority and crew communication must agree.
Air	Control surfaces or rotor thrust affect attitude and forces; power and configuration interact with energy.	Attitude, airspeed, altitude and flight path differ from geographic position and ground track.
Water	Thrust and turning effects depend on propulsion architecture and water flow; current and wind add drift.	Heading, track, speed through water, speed over ground, depth reference and clearance are separate.
Space	Attitude change is not necessarily trajectory change; propulsion changes velocity in a stated frame.	Estimated position and velocity, pointing, timing, resource state and target-relative motion.

A command is not an observation

Distinguish **requested** state, **indicated** state, **estimated** state and actual physical state. A selected heading does not show the achieved track. A map line does not prove clearance. A mode label does not guarantee that the mode is available or engaged. The card should record how uncertainty is noticed, not silently assume perfect sensors.

A new phase changes the problem

Wheels-to-water, hover-to-wing-borne flight, hull-to-foil support and orbit-to-entry can change what inputs do and what recovery remains possible. A story or interface that keeps the same controls should still explain any underlying remapping or automation. If that explanation is absent, label the gap rather than inventing a mechanism.

Researching an omitted family: identify its medium, support, propulsion, control mapping, feedback, phase transitions and support organization. Find current manufacturer or operator documents and applicable authority. Then record what remains unknown and what training is still needed.

People automation and access

Roles are functions not seat labels

Separate mission authority, operating task, supervision, maintenance, passenger care and recovery support. One person may hold several roles only when their competence, workload and physical location allow it. A train passenger, ferry passenger or spacecraft visitor does not become an operator by understanding an instrument display.

Crew coordination matters during simultaneous tasks: road traffic observation and passenger assistance, sail handling and helm work, aircraft flight and communication, or submersible control and surface recovery. Identify handoffs, unavailable personnel and conflicting commands before treating a role list as a working operating system.

Automation must be described by capability

Do not equate assistance with a driverless service. NHTSA's driver-assistance distinctions retain driver engagement for assistance functions. More automated systems depend on the actual approved system, operational design domain and jurisdiction; responsibilities cannot be inferred from a marketing name. This reference does not repeat blanket claims about current market availability. [P1]

Orion publicly describes automated and manual modes and commands from crew and ground. Perseverance describes bounded onboard route choice within ground-team goals. Neither example justifies assuming every automated system has the same handover, override or supervision model. [S8, S9]

Record what automation senses, commands and cannot judge; what mode is active; who has decision authority; and what happens when communication or a required capability is unavailable. Do not assume human intervention is always possible, necessary or safe. Proficiency includes understanding automation limits rather than treating it as a substitute for system knowledge. [F2]

Individual access and task fit

Assess boarding and transfer, reach, seating and restraints, sensory information, language and cognitive load, controls, mobility aids and exit paths. Normal access and emergency access are different questions. Disability does not establish incompetence; the actual task and interface determine what adaptation, training or assistance is needed.

NHTSA recommends qualified driver-rehabilitation assessment and appropriate modifications for adapted vehicles. Powered wheelchair guidance provides a separate device context. Neither supports assuming foot pedals, two-handed steering or a single sensory profile for all operators. [L28, L29]

Concurrency counterexample: an expert cannot simultaneously occupy two separated stations and provide hands-on assistance at a blocked exit. Expertise does not create another person or an accessible path. A proposed solution must be supported, not assumed.

Training maintenance and evidence limits

Reading and qualification are different

A conceptual map can explain why controls differ. It cannot establish hands-on skill, medical or task suitability, current type familiarity, route qualification or permission. Supervised practice, simulation where appropriate, assessment and recurrent proficiency belong to the relevant training system. Reading a handbook or succeeding in an entertainment game is not a license.

Road and workplace requirements depend on jurisdiction and use. California licensing and US FAA certification are examples, not worldwide rules. Commercial seafaring has international STCW training and watchkeeping standards implemented through applicable authorities; recreational boating is a different regulatory context. [L04, L06, A8, A9, L38]

Readiness is sustained work

Inspection, maintenance release, loading, weather, infrastructure, consumables, spares and trained support affect whether a capability can actually be used. Maintenance after a defect does not remove the need for applicable pre-use inspection. Operator skill does not confer authority or competence for repair, high-voltage isolation, pressure systems, lifting or rescue. [P2, L12, L17]

Space systems add tracking networks, power and thermal management, validated software and sequences, long-duration resources and sometimes no practical physical repair access. A communications outage does not suspend onboard duties or life support; relays do not erase the finite travel time of signals. [S2, S10, S11, P3]

Evidence hierarchy and remaining work

Use the exact model's current approved manual and supplements, competent instruction and applicable rules for real action. Manufacturer pages can support their own design, not fleet-wide guarantees. Investigations show particular failure mechanisms, not universal operating prescriptions. Fiction summaries support story premises, not physical proof.

This research checked selected official government, regulator, training-body, operator, manufacturer and fictional-continuity sources. It did not read every complete manual or reconcile every statute. Some large FAA chapters and the California commercial handbook exceeded retrieval size; catalogue, accessible pages and indexed official excerpts were used. The 1942 airship manual is historical. Older officially hosted documents still require applicability checks.

Open boundary: no physical inspection, current model-specific operating determination, hands-on competence assessment or complete canon reconciliation was performed. The internal desk reviews below assess this scoped dossier, not the whole Blackwish project or any vehicle's safety.

Fictional vehicles and evidence

Keep **world scope** and **support status** separate. A claim may concern the actual world, a named fictional continuity or an interface representation. Its support may be a source, an inference, an explicit premise or an unknown. Internal consistency does not demonstrate physical feasibility, safety or operating permission.

Four official continuity examples

The following narrow Star Wars Databank examples illustrate source discipline across domains. They are not a franchise survey, technical manual or complete canon reconciliation.

Example	Sourced story premise	Do not infer
Land AT-AT walker	A four-legged transport used by Imperial ground forces. [F4]	Detailed leg controls, stability envelope, maintenance sequence or safe operation.
Air Snowspeeder	T-47 craft adapted for Hoth's cold; a pilot and a distinct rear-facing crew role. [F5]	Unrestricted cold-weather capability or one universal cockpit.
Water and air Wookiee catamaran	A twin-hulled craft with engines and fictional repulsorlifts has water and air mobility. [F6]	A real catamaran can fly, or transition controls and resource limits are established.
Space Millennium Falcon	A modified freighter with recurring hyperdrive unreliability. [F7]	A demonstrated faster-than-light mechanism or guaranteed reliability.

Preserve the status of each mechanism

Distinguish demonstrated technology in a specified application, engineering extrapolation, unresolved proposal, invented mechanism and unexplained in-story phenomenon. These are evidence labels, not grades of literary quality. Magic need not receive a pseudo-scientific explanation; real terminology must not launder an invention into proven physics.

NASA describes solar-electric propulsion as using electrical power to ionize and accelerate propellant. Calling a fictional drive an ion drive does not establish unlimited propellant, arbitrary acceleration or cross-domain capability. Record what the story actually grants and what remains unknown. [F1]

Canon change boundary: a proposed repair is not accepted canon. Keep source-supported story facts, inferred details and suggested revisions in separate fields. Unknown, not applicable and none known are legitimate entries.

Synthetic journeys across media

All examples on this page are original hypothetical prompts, not claims about an existing franchise. They show how the same research method can support different genres without making every vehicle obey the same fictional physics.

Domain	Synthetic example	Useful consistency question
Land	A walking archive in a fantasy mystery	Does its established leg clearance permit the route, and who can authorize it to move?
Air	A wind-bound sky-barge in an adventure	Does an established open-sky enchantment still work inside a roofed cavern, or is an exception explicitly introduced?
Water	A ferry in a realist romance or horror story	Who handles the boat while another person assists a passenger, and is rescue reachable or none known?
Space	An orbital courier in procedural science fiction	Which state is being changed: pointing, orbit or target-relative motion? Which drive properties are invented?

A scene sequence not a startup checklist

Establish the journey's purpose and limitation; identify authority and roles; establish readiness, access and resources; depict departure as a meaningful transition; show an input, response and constraint during travel; introduce changed information; show a supported adjustment, recovery attempt or unrecoverable outcome; then establish the end state and consequences.

Not every beat belongs on the page or screen. Offscreen work may remain implied when it does not contradict established constraints. Omission is not automatically inconsistency, and intentional tragedy must not be rewritten into a falsely safe procedure.

Book film and game

Book: viewpoint can show interpretation, uncertainty and workload; a narrator's belief is not necessarily world fact. **Film:** framing, sound, indications and performance can preserve causal links through compressed time. An edit alone does not repair damage or replenish resources.

Game: separate player commands from character action and crew automation. Route markers, simplified controls, pause, checkpoints, respawn and time compression may be interface rules rather than in-world powers. A playable model is not therefore a validated training simulator.

Adaptation invariants: compare authority, knowledge, resources, constraints, failure costs and end state. Deliberate changes are allowed; record them as version changes rather than quietly treating a medium shortcut as a new vehicle capability.

Vehicle card Identity and operation

Copy this card for an actual type or a fictional vehicle. Fill only what evidence establishes. Use **unknown**, **not applicable** or **none known** rather than invented precision. A completed card is not a certification.

Field	Record
1 Identity and purpose	Model, version or continuity; era; domains; mission; passenger or cargo role.
2 Claim ledger	Exact claim; world scope; source or premise; inference; limits; conflict; open question.
3 Environment	Medium, weather, terrain or depth, clearance, infrastructure and phase transitions.
4 Motion and support	Support, propulsion, turning, speed or state change, environmental coupling, fictional overrides.
5 Resources	Energy, propellant, consumables, heat, storage, replenishment and evidenced reserves.
6 Authority	Who sets the mission, authorizes movement, changes route, accepts risk or cancels.
7 Crew and roles	Onboard, remote or autonomous roles; competence; concurrent work; relief and handoffs.
8 Control and automation	Input mapping, modes, feedback, limits, supervision, handover and conflicting commands.

Source and assumption notes

For each consequential answer, record its source identifier or explicit fictional premise, model or continuity version, date checked and applicability limit. Label an inference as an inference. Do not treat another unsourced inference as independent evidence.

Authority reminder: possession of a seat, key, control or login is not the same as permission to select the mission, accept risk or operate. Research ratification is not action authorization.

Vehicle card Support and review

Continue the same vehicle record. Recovery can be unknown or none known; accessibility, training and evidence gaps must remain visible.

Field	Record
9 Navigation and links	Position and frame, maps and sensors, uncertainty, freshness, delay and route permission.
10 Access	Boarding, transfer, reach, restraints, displays, assistance and normal versus emergency exit.
11 Readiness	Inspection, wear, faults, qualified repair, tools, spares, downtime and release assumptions.
12 End state and recovery	What stop, hold, land, moor or safe mode means; reachability, evacuation or none known.
13 Failure and consequences	Coupled losses, claimed independence, irreversible outcomes and recovery limits.
14 Medium translation	What is perceived, automated, offscreen or compressed; interface versus world rules.
15 Review and boundary	Sources, training gaps, tests, contradictions, proposals, uncertainties and ratification state.

Compact claim: Under conditions C, actor A in role R can request action X through interface I, with feedback F and limits L. Fallback is B, unknown or none known. This is a proposition to examine, not operating authorization.

Open questions and training gaps

What additional model-specific evidence is needed? Which claims conflict? What cannot be established from desk research? Which instruction, assessment, access adaptation or separate authorization would be required before any proposed real action?

Disposition: source-supported / inferred / hypothetical / unknown. Record proposed repairs separately from accepted text. A complete card is not proof that a vehicle, route or operator is safe or qualified.

Zen reviews one and two

Three text-only reviews apply POINT, BRIDGE, PATH and SHADOW to this dossier. POINT states the claim or objective; BRIDGE checks evidence and authority; PATH examines consequences, alternatives and termination; SHADOW seeks blind spots and counterexamples. These are scoped reviews, not full-canon verification or empirical tests.

Review one Epistemic separation

POINT: a sourced fictional premise, actual-world mechanism and interface shortcut must remain distinguishable at claim level.

BRIDGE: real-domain statements carry official source identifiers. Databank entries support narrow story facts only; original examples are labeled hypothetical. Historical airship material and the Mars concept paper are distinguished from present operational evidence.

PATH: trace a capability to its source, version, assumptions and limits. Reject the inference that a named ion drive grants unlimited resources. Where source scope is insufficient, narrow the claim or preserve the gap rather than promote it to fact.

SHADOW: unexplained magic, unreliable narration and mystery may intentionally remain unresolved. A blank is not automatically a contradiction. **Finding:** the dossier separates claim types, but no complete Blackwish canon or full-manual audit was available; those remain outside the conclusion.

Review two Operational coherence and access

POINT: controls alone do not constitute an operating system; roles, information, resources, maintenance and recovery must be coherent.

BRIDGE: compare real mechanisms and phases, then use the vehicle card for a stated fictional or actual-type claim. No score from that card certifies safety. Access is evaluated against tasks and interfaces rather than assumptions about disability.

PATH: consider normal travel, degraded information, an unavailable operator, one domain transition and one passenger-assistance need. Check simultaneous work and actual recovery links. Alternatives may include a supported crew arrangement, narrower capability or no known recovery.

SHADOW: a single expert cannot occupy separated stations and assist at an exit simultaneously. A visible route need not fit the loaded craft. **Finding:** the universal-operator and route-equals-safety claims fail these hypothetical counterexamples; no replacement crew or rescue capability is invented.

Review disposition: scoped reasoning completed; identified overgeneralizations are narrowed in this reference. Remaining gaps are recorded as unknown or outside scope, not silently marked passed.

Zen review three and Death Knell

Review three Adaptation and narrative economy

POINT: preserve relevant causal and ethical consequences across media, not necessarily identical controls or elapsed screen time.

BRIDGE: the scene sequence and medium ledger connect book interpretation, film cues and game interaction. They are editorial tools, not engineering evidence.

PATH: compare three hypothetical versions of a damaged ferry journey. In a book, damage persists; a film cut can omit repair work without proving it happened; a game respawn may be an interface convenience. Preserve damage consequences or explicitly record a changed continuity.

SHADOW: exhaustive detail can harm pacing. Comedy, ambiguity and intentional rule changes are permissible. **Finding:** medium compression alone cannot establish repair, rescue or competence; a few decisive cues can preserve consistency without a technical lecture.

Bounded Death Knell adversarial audit

Target claim: the displayed route is visible and autopilot is engaged, therefore every passenger can reach safety. **Scope:** a synthetic scene with only those premises. No live vehicle, deployment or human experiment is involved.

Chosen counterexample	Result and correction
The route passes under an arch narrower than the craft.	Displayed route does not prove traversability. Add supported clearance evidence or narrow the claim.
Arrival occurs but an occupied exit is inaccessible.	Navigation success does not establish accessible evacuation. Record the access gap.
A distress message is sent but no rescuer can reach the vehicle.	Communication is not rescue availability. Recovery remains none known if that is all the premises support.

Outcome: the target implication is contradicted within the stated hypothetical model. Reject or narrow that claim, not the entire project. An underspecified claim stays open; no contradiction found would mean only no contradiction found in the bounded cases. Intentional unsafe or tragic depiction may stand, without a false safe-operation label.

Stopping condition reached: the selected target has documented counterexamples and correction options. No exhaustive consistency, scientific validation, vehicle safety or training claim follows. Proposed revisions remain unratified; any real operation or deployment also needs separate explicit authorization and applicable qualifications.

Final disposition: RESEARCH / REVIEWED / NOT RATIFIED. The three reviews and bounded audit concern this educational dossier only.

Sources and applicability

All listed sources were checked on 22 October 2026. Dates below describe publication or update only where established. Otherwise no publication date is asserted. A check date does not make older guidance current for every operation.

[L01] Caterpillar controls

https://www.cat.com/en_US/articles/for-owners/excavator-joystick-controls.html/

[L02] Alvin systems

https://ndsf.whoi.edu/alvin/systems/

[L03] NHTSA bicycle safety

https://www.nhtsa.gov/road-safety/bicycle-safety

[L04] California motorcycle handbook, 2024 PDF, especially printed pp. 11-15

2024 handbook; selected relevant material, including printed pages 11-15.

https://qr.dmv.ca.gov/portal/file/motorcycle-driver-handbook-pdf/

[L05] California driver introduction

https://www.dmv.ca.gov/portal/handbook/california-driver-handbook/introduction-to-driving/

[L06] California commercial driver information

https://www.dmv.ca.gov/portal/driver-education-and-safety/special-interest-driver-guides/truck-drivers/commercial-driver-information/

[L07] commercial handbook PDF

Full PDF exceeded retrieval size; accessible DMV pages and indexed material were used.

https://www.dmv.ca.gov/portal/file/california-commercial-driver-handbook-pdf/

[L08] FRA expert-engineer study

https://railroads.dot.gov/sites/fra.dot.gov/files/2021-08/Expert%20Locomotive%20Engineer%20Mental%20Model.pdf

[L09] FRA PTC

https://railroads.dot.gov/research-development/program-areas/train-control/ptc/positive-train-control-ptc

Source register continued

All listed sources were checked on 22 October 2026. Dates below describe publication or update only where established. Otherwise no publication date is asserted. A check date does not make older guidance current for every operation.

[L10] HSE ATV families

https://www.hse.gov.uk/agriculture/topics/machinery/farm-vehicles-3.htm

[L11] HSE tractor stability

https://www.hse.gov.uk/agriculture/topics/machinery/farm-vehicles-2.htm

[L12] HSE safe machinery use

https://www.hse.gov.uk/agriculture/topics/machinery/safe-use-1.htm

[L13] NIOSH skid-steer alert

https://www.cdc.gov/niosh/publications/numbered/2011-128.html

[L14] NPS kayak/boat safety

https://www.nps.gov/kefj/planyourvisit/kayak-and-boat-safety.htm

[L15] NPS guide skills task book

https://www.nps.gov/piro/upload/PIRO_Kayak_Guide_Task_Book_v2.pdf

[L16] RYA points of sail

https://www.rya.org.uk/training/do-you-know-your-points-of-sail/

[L17] Transport Canada predeparture/maintenance

https://tc.canada.ca/en/marine-transportation/preparing-operate-your-vessel/maintaining-safe-pleasure-craft

[L18] NOAA Coast Pilot

https://nauticalcharts.noaa.gov/publications/coast-pilot/

[L19] USCG navigation rules

https://www.navcen.uscg.gov/navigation-rules-amalgamated

[L20] Candela foiling mechanism

Manufacturer example for its own system, not all hydrofoils.

https://candela.com/technology/foiling-system/

Source register continued

All listed sources were checked on 22 October 2026. Dates below describe publication or update only where established. Otherwise no publication date is asserted. A check date does not make older guidance current for every operation.

[L21] UK Hovercraft Code landing page

Officially hosted older code; applicability must be checked for the actual operation.

https://www.gov.uk/government/publications/the-hovercraft-code-of-practice-cop-23

[L22] UK code PDF

https://assets.publishing.service.gov.uk/media/5ed0dda4e90e0754d5088ad5/Hovercraft_Code_of_Practice__COP_24__-_Tagged.pdf

[L23] IMO HSC

https://www.imo.org/en/ourwork/safety/pages/hsc.aspx

[L24] WHOI Alvin operations

https://ndsf.whoi.edu/alvin_operations/

[L25] Alvin specifications

https://ndsf.whoi.edu/alvin/specifications/

[L26] Horse-drawn vehicle code

https://www.gov.uk/government/publications/code-of-practice-for-horse-drawn-vehicles/code-of-practice-for-horse-drawn-vehicles

[L27] equine welfare

https://www.gov.uk/government/publications/code-of-practice-for-the-welfare-of-horses-ponies-donkeys-and-their-hybrids

[L28] NHTSA adapted vehicles

https://www.nhtsa.gov/vehicle-safety/adapted-vehicles

[L29] NHS powered-wheelchair guide

https://www.berkshirehealthcare.nhs.uk/media/109513718/powered-wheelchairs-guide.pdf

[L30] Cableway guidance

https://www.gov.uk/government/publications/clarifying-information-for-the-cableway-installations-regulations-2018

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All listed sources were checked on 22 October 2026. Dates below describe publication or update only where established. Otherwise no publication date is asserted. A check date does not make older guidance current for every operation.

[L31] lifts guidance

https://www.gov.uk/government/publications/lifts-regulations-2016/lifts-regulations-2016-great-britain

[L32] NTSB Seattle investigation

https://www.ntsb.gov/investigations/pages/HWY15MH011.aspx

[L33] NTSB marine safety priorities

https://www.ntsb.gov/advocacy/mwl/pages/mwl-21-22/mwl-ms-01.aspx

[L34] US Forest Service safety/avalanche guide

https://www.fs.usda.gov/Internet/FSE_DOCUMENTS/stelprdb5447203.pdf

[L35] RYA equipment guide

https://www.rya.org.uk/water-safety/equipment-for-uk-pleasure-vessels/

[L36] RYA electronic-navigation syllabus

https://www.rya.org.uk/media/sokklzc4/5b2_mut11723198255604924966.pdf

[L37] C-8 owner's manual

https://help.candela.com/media/manual_c8_en.pdf

[L38] IMO STCW

https://www.imo.org/en/ourwork/humanelement/pages/stcw-conv-link.aspx

[A1] FAA Pilot’s Handbook of Aeronautical Knowledge

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/phak

https://www.faa.gov/sites/faa.gov/files/18_phak_ch16.pdf

[A2] FAA Airplane Flying Handbook

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/airplane_handbook

[A3] FAA Helicopter Flying Handbook

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/helicopter_flying_handbook

Source register continued

All listed sources were checked on 22 October 2026. Dates below describe publication or update only where established. Otherwise no publication date is asserted. A check date does not make older guidance current for every operation.

[A4] FAA handbook catalogue (gyroplane-specific warning and historical manual dates)

Catalogue distinguishes gyroplane guidance from current helicopter guidance.

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation

[A5] FAA Glider Flying Handbook

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/glider_handbook

[A6] FAA Balloon Flying Handbook

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/Balloon_Flying_Handbook

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/Balloon_Flying_Handbook_FAA-H-8083-11B/bfh_chapter_7.pdf

[A7] FAA historical Airship Pilot Manual

Historical 1942 manual. Principles only; not a current operating checklist.

https://www.faa.gov/sites/faa.gov/files/regulations_policies/handbooks_manuals/aviation/airship_pilot_manual.pdf

[A8] FAA Remote Pilot Study Guide

https://www.faa.gov/sites/faa.gov/files/regulations_policies/handbooks_manuals/aviation/remote_pilot_study_guide.pdf

https://www.faa.gov/uas/commercial_operators/become_a_drone_pilot

[A9] FAA powered-lift FAQ

https://www.faa.gov/air-taxis/FAQ

https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1042955

[A10] FAA Seaplane/Skiplane/Float-Ski Helicopter Handbook

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/seaplane_handbook

[S1] NASA rocket thrust

https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/rocket-thrust-equation/

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All listed sources were checked on 22 October 2026. Dates below describe publication or update only where established. Otherwise no publication date is asserted. A check date does not make older guidance current for every operation.

[S2] NASA spaceflight operations

https://www.nasa.gov/reference/jsc-spaceflight-operations/

[S3] NASA spacecraft navigation

https://science.nasa.gov/learn/basics-of-space-flight/chapter13-1/

[S4] NASA/JPL reference frames

https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/req/frames.html

[S5] NASA onboard pointing/control systems

https://science.nasa.gov/learn/basics-of-space-flight/chapter11-2/

[S6] NASA Orion systems

https://www.nasa.gov/missions/meet-nasas-orion-spacecraft/

[S7] NASA Shuttle aeronautics

https://www.nasa.gov/centers-and-facilities/langley/the-aeronautics-of-the-space-shuttle/

[S8] NASA Orion avionics/software

https://www.nasa.gov/reference/avionics-and-software/

[S9] NASA rover autonomy/ground role

2021 account of Perseverance AutoNav and ground-team role.

https://www.nasa.gov/solar-system/nasas-self-driving-perseverance-mars-rover-takes-the-wheel/

[S10] NASA spacecraft power

https://science.nasa.gov/learn/basics-of-space-flight/chapter11-3/

[S11] NASA ECLSS

https://www.nasa.gov/reference/environmental-control-and-life-support-systems-eclss/

Source register continued

All listed sources were checked on 22 October 2026. Dates below describe publication or update only where established. Otherwise no publication date is asserted. A check date does not make older guidance current for every operation.

[P1] NHTSA Driver Assistance Technologies

Used for assistance distinctions, not its potentially stale market-availability statement.

https://www.nhtsa.gov/vehicle-safety/driver-assistance-technologies

[P2] FAA Advanced Preflight After Maintenance

Page updated 4 February 2022.

https://www.faa.gov/newsroom/safety-briefing/advanced-preflight-after-maintenance

[P3] NASA Mars Communications Disruption and Delay

2023 architecture concept-review whitepaper. Not demonstrated crewed Mars operations.

https://www.nasa.gov/wp-content/uploads/2024/01/mars-communications-disruption-and-delay.pdf

[F1] NASA Solar Electric Propulsion

Actual-world factual anchor, not evidence for fictional drive performance.

https://www.nasa.gov/space-technology-mission-directorate/tdm/solar-electric-propulsion/

[F2] FAA CFIT and Automation Overreliance

Actual-world factual anchor for automation limits and proficiency.

https://www.faa.gov/newsroom/safety-briefing/cfitautomation-overreliance

[F3] NASA Gravity and Mechanics

Actual-world factual anchor for motion after propulsion ends.

https://science.nasa.gov/learn/basics-of-space-flight/chapter3-2/

[F4] Star Wars Databank AT-AT Walker

Fictional continuity source only.

https://www.starwars.com/databank/at-at-walker

[F5] Star Wars Databank Snowspeeder

Fictional continuity source only.

https://www.starwars.com/databank/snowspeeder

[F6] Star Wars Databank Wookiee Catamaran

Fictional continuity source only.

https://www.starwars.com/databank/wookiee-catamaran

[F7] Star Wars Databank Millennium Falcon

Fictional continuity source only.

https://www.starwars.com/databank/millennium-falcon